Marmara University
Electrical and Electronics Engineering
Probability and Statistics
Problem Sets with Solutions #3

[1] A point x is chosen at random within the interval (0, 1). If it is known that $x \geq \frac{1}{2}$, what is the probability that $x \geq 7/8$.

SOLUTION: $P[x \geq \frac{7}{8} \mid x \geq \frac{1}{2}] = \frac{P[x \geq \frac{7}{8} \cap x \geq \frac{1}{2}]}{P[x \geq \frac{1}{2}]} = \frac{P[x \geq \frac{7}{8}]}{P[x \geq \frac{1}{2}]} = \frac{1/8}{1/2} = \frac{1}{4}$ ☒

[2] An urn contains 3 red balls and 2 black balls. If two balls are chosen without replacement, find the probability that the second ball is black, if it is known that the first ball chosen is black.

SOLUTION: $B_1 = \{$First ball chosen is black$\}$, $B_2 = \{$Second ball chosen is black$\}$

$$P[B_2 \mid B_1] = \frac{P[B_2 \cap B_1]}{P[B_1]} = \frac{\frac{2}{5} \cdot \frac{1}{4}}{\frac{2}{5}} = \frac{2}{8} = \frac{1}{4}$$ ☒

[3] A coin is tossed 11 times in succession. Each 11-tuple outcome is equally likely to occur. If the first 10 tosses produad all heads, what is the probability that the 11th toss will also be a head?

SOLUTION: $B = \{$The first 10 tosses produad all heads$\}$
$A = \{$The 11th toss is head$\}$

$$P[A \mid B] = \frac{P[A \cap B]}{P[B]} = \frac{P[11 \text{ heads}]}{P[10 \text{ heads}]} = \frac{(\frac{1}{2})^{11}}{(\frac{1}{2})^{10}} = \frac{1}{2}$$

[4] A new bus runs on Sunday, Tuesday, Thursday and Saturday, while an older bus runs on the other days. The new bus has a probability of being on time of 2/3 while the olde bus has a probability of only 1/3. If a passanger chooses an arbitrary day of the week to ride the bus, what is the probability that the bus will be on time

Solution: $P[\text{on time}] = P[\text{on time/new bus}]\, P[\text{new bus}] + P[\text{on time/old bus}] \cdot P[\text{old bus}]$

$$= \frac{2}{3} \cdot \frac{4}{7} + \frac{1}{3} \cdot \frac{3}{7}$$

$$= \frac{11}{21}$$ ☒

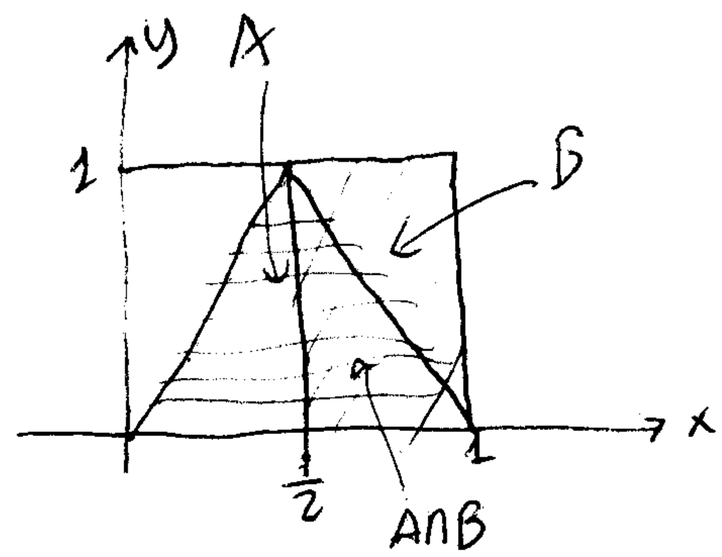
[5] A sample space is given by $S=\{(x,y): 0\le x\le 1,\ 0\le y\le 1\}$ Determine $P[A|B]$ for the events

$A=\{(x,y): y\le 2x,\ 0\le x\le \frac{1}{2} \text{ and } y\le 2-2x,\ \frac{1}{2}\le x\le 1\}$

$B=\{(x,y): \frac{1}{2}\le x\le 1,\ 0\le y\le 1\}$

Are A and B independent?

SOLUTION:



$P[A]=\frac{1\cdot 1}{2}=\frac{1}{2}$

$P[B]=\frac{1}{2}$

$P[A\cap B]=\frac{1}{4}$

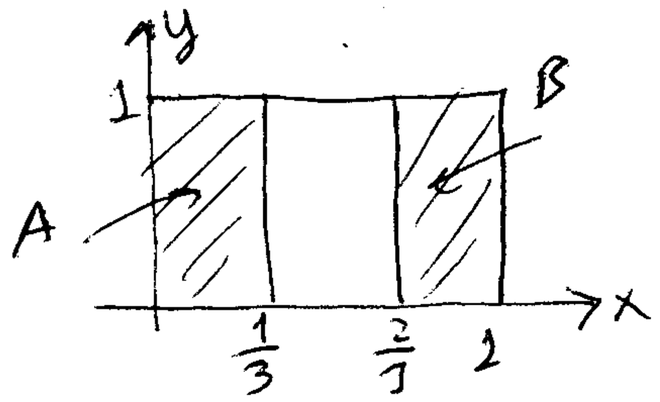
$P[A|B]=\frac{P[A\cap B]}{P[B]}=\frac{\frac{1}{4}}{\frac{1}{2}}=\frac{1}{2}$

$P[A\cap B]=\frac{1}{4}=P[A].P[B]=\frac{1}{2}\cdot\frac{1}{2}$

A and B are independent. ☑

[6] Give an example of two events that are mutually exclusive but not independent.

SOLUTION:



$A\cap B=\phi$ (A and B are mutually exclusive)

$P[A]=P[B]=\frac{1}{3}$

$P[A\cap B]=0\neq P[B].P[A]=\frac{1}{9}$

A and B are not independent ☑

[7] It is known that if it rains, there is a 50% chance that a sewe will overflow. Also, if the sewe overflows, there is a 30% chance that the road will flood. If there is a 20% chance that it will rain, what is the probability that the road will flood?

SOLUTION: Let $A_3=\{\text{road floods}\}$, $A_2=\{\text{sewe overflows}\}$, $A_1=\{\text{it rains}\}$

Given probability is $P[A_3]=P[A_3|A_2,A_1].P[A_2|A_1].P[A_1]$

$=P[A_3|A_2].P[A_2|A_1].P[A_1]$ (flood depends only on sewer overflow)

$=(0.3)(0.5)(0.2)=0.03$ ☑

[8] Consider the sample space $S=\{1,2,3,4\}$ Each simple event is equally likely Let, $A=\{1,2\}$, $B=\{1,3\}$, $C=\{1,4\}$. Are these events independent?

SOLUTION.

$P[A\cap B]=P[1]=\frac{1}{4}=P[A].P[B]=\frac{1}{2}\cdot\frac{1}{2}$

$P[A\cap C]=P[1]=\frac{1}{4}=P[A].P[C]=\frac{1}{2}\cdot\frac{1}{2}$

$P[B\cap C]=P[1]=\frac{1}{4}=P[B].P[C]=\frac{1}{2}\cdot\frac{1}{2}$

$P[A\cap B\cap C]=P[1]=\frac{1}{4}\neq P[A].P[B].P[C]=\frac{1}{2}\cdot\frac{1}{2}\cdot\frac{1}{2}$

} They are not independent (but pair wise independent!)

☑

[9] A drunk person wunders aimlessly along a path going one step forward with probability 1/2 and going backward one step with probability 1/2. After 10 steps, what is the probability that he has moved 2 steps forward?

SOLUTION: Let a and b be the numbe of steps forward and backward, respectively.

Then,
$$\left.\begin{aligned} a+b &= 10 \\ a-b &= 2 \end{aligned}\right\} \Rightarrow \begin{aligned} a &= 6 \\ b &= 4. \end{aligned}$$

If we consider one step forward as "success" and one step backward as a "failure", then

$$P(6 \text{ successes in } 10 \text{ trials}) = \binom{10}{6} (P[\text{success}])^6 \cdot [P[\text{failure}])^{10-6}$$

$$= \binom{10}{6} \left(\frac{1}{2}\right)^6 \left(\frac{1}{2}\right)^4 = \frac{10.9.8.7}{4.3.2.1} \left(\frac{1}{2}\right)^{10} = \frac{105}{512}$$ ☑

[10] Prove that the geometric probability law give by

$$P[k] = (1-p)^{k-1} \cdot p, \quad k = 1, 2, \ldots$$

is a valid probability assignment.

SOLUTION. If it is a valid probability assignment, the sum of all probabilities must be equal to 1.

$$\sum_{k=1}^{\infty} P[k] = \underbrace{\sum_{k=1}^{\infty} (1-p)^{k-1} \cdot p}_{\text{let } u = k-1} = p \sum_{u=0}^{\infty} (1-p)^u = p \cdot \frac{1}{1-(1-p)} = p \cdot \frac{1}{p} = 1$$ ☑

[11] An urn contains red, black and white balls. The proportion of red is 0.4, the proportion of black is 0.4 and the proportion of whit is 0.2. If 5 balls are drawn with replacement, what is the probability of 2 red, 2 black and 1 whit in any order?

SOLUTION

$$P[\text{red}] = p_1 = 0.4$$
$$P[\text{black}] = p_2 = 0.4$$
$$P[\text{white}] = p_3 = 0.2$$

$$P[2 \text{ red}, 2 \text{ black}, 1 \text{ white}] = \frac{5!}{2!\,2!\,1!} p_1^2 \cdot p_2^2 \cdot p_3$$

$$= 30 (0.4)^2 \cdot (0.4)^2 \cdot (0.2)$$

$$= 0.1536$$

☑

[12] We derive the multinomial coefficient for $N=3$. This will yield the number of ways that an M-tuple can be formed using k_1 1's, k_2 2's and k_3 3's. To do so, we choose k_1 places in the M-tuple for the 1's. There will be $M-k_1$ positions remaining. Of these positions, choose k_2 places for the 2's. Fill in the remaining $k_3 = M-k_1-k_2$ positions using the 3's. Using this result, determine the number of M-digit sequences with k_1 1's, k_2 2's and k_3 3's.

Solution:

$$\binom{M}{k_1}\binom{M-k_1}{k_2} = \frac{M!}{(M-k_1)!\,k_1!}\,\frac{(M-k_1)!}{(M-k_1-k_2)!\,k_2!} = \frac{M!}{k_1!\,k_2!\,(M-k_1-k_2)!} = \frac{M!}{k_1!\,k_2!\,k_3!}$$

$M-k_1-k_2=k_3$

Hence, the number of M-digit sequences with k_1 1's, k_2 2's and k_3 3's is

$$\binom{M}{k_1\,k_2\,k_3} = \frac{M!}{k_1!\,k_2!\,k_3!}$$ [5]

[13] Show that the multinomial probability law reduces the binomial law for $N=2$

Solution:

Multinomial probability law $P(k_1,k_2,\cdots,k_N) = \frac{M!}{k_1!\,k_2!\cdots k_N!}P_1^{k_1}P_2^{k_2}\cdots P_N^{k_N}$ $(k_1+k_2+\cdots+k_N=M)$, $P_1+P_2+\cdots+P_N=1$

Binomial law $P(k) = \binom{M}{k}p^k(1-p)^{M-k}$

Let $N=2$ in the multinomial probability law. Then

$$P(k_1,k_2) = \frac{M!}{k_1!\,k_2!}P_1^{k_1}P_2^{k_2} \qquad k_1+k_2=M \Rightarrow k_2=M-k_1,\quad P_1+P_2=1$$

$$= \frac{M!}{k_1!\,(M-k_1)!}P_1^{k_1}P_2^{M-k_1}$$

$$= \binom{M}{k_1}P_1^{k_1}P_2^{M-k_1}$$

Let $k_1=k$ and $P_1=P$, we have

$$P(k,M-k) = \binom{M}{k}p^k(1-p)^{M-k}$$

which is recognized as binomial law. ∎